

System Power-Up

init()

```
1. Clock_Enable(GPIOA_BUS); Clock_Enable(GPIOB_BUS); Clock_Enable(GPIOC_BUS); Clock_Enable(TIM3_BUS);  
2. GPIO_InitPinsSameMode(GPIOA, {2,10}, INPUT, PULL_DOWN); GPIO_InitPin(GPIOB, 5, INPUT, PULL_DOWN); // New Item, Step Gate, E-Stop  
3. GPIO_InitPinsSameMode(GPIOA, {9,0,1,4,3,11,8}, OUTPUT, PULL_DOWN); // Fault LED + Real RGB + Sensed RGB  
4. GPIO_InitPin(GPIOB,0,ALT,NONE); GPIO_InitPin(GPIOC,7,ALT,NONE); // servo pins -> alt-function mode  
5. GPIO_ConnectPeripheral(GPIOB,0,1); GPIO_ConnectPeripheral(GPIOC,7,1); // AF1 = TIM3  
6. Timer_Init(TIM3, TIM3_IRQn, psc=47, arr=19999, ...); // both gates share ONE timer, 50 Hz  
7. PWM_Init(TIM3, CH2, PWM1, true); PWM_Init(TIM3, CH3, PWM1, true); // CH2=Gate1(PC7), CH3=Gate2(PB0); starts counter  
8. Gate1_SetWidth(GATE_ANGLE_NEUTRAL); Gate2_SetWidth(GATE_ANGLE_NEUTRAL); // both = 1500  
9. EXTI_Init(GPIOA,2,RISING,EXTI2_3_IRQn); EXTI_Init(GPIOA,10,RISING,EXTI4_15_IRQn); EXTI_Init(GPIOB,5,RISING,EXTI4_15_IRQn);  
10. SysTick_Init();  
typedef enum {STATE_IDLE, STATE_ITEM_NEW, STATE_STAGE1, STATE_STAGE2, STATE_FAULT} SystemState_t;  
volatile SystemState_t current_state = STATE_IDLE; volatile bool isStateChange = false;
```

ISR 1 (New Item): EXTI2_3_IRQHandler

EXTI->RPR1 & (1<<2)?

NO

Exit ISR

YES

EXTI->RPR1 |= (1<<2); (clear flag — always runs)

current_state == STATE_IDLE?

YES

changeState(STATE_ITEM_NEW);

Exit ISR

ISR 2 (Step Gate & E-Stop): EXTI4_15_IRQHandler

shared vector — Step Gate (bit10) checked first, then E-Stop (bit5), independently

EXTI->RPR1 & (1<<10)? (Step Gate)

YES

switch(current_state)

STAGE1

ITEM_NEW

STAGE2

NO

STAGE1:
changeState(STAGE2);

ITEM_NEW:
changeState(STAGE1);

default: no change

STAGE2:
changeState(IDLE);

EXTI->RPR1 |= (1<<10); (only if bit10 was pending)

EXTI->RPR1 & (1<<5)? (E-Stop — checked independently)

NO

Exit ISR

YES

changeState(STATE_FAULT);
EXTI->RPR1 |= (1<<5);